

A/B Testing & Randomized Experiments

ECON 4370 / BANA 4373 — Applied Data Tools for Economics & Business

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Class 1

The Logic of Randomization

From Expert Opinion to Experimental Evidence

Before We Start — A Quick Poll

Two versions of a Netflix thumbnail for the same movie



Source: Netflix Tech Blog, 2017

Version A



Source: Netflix Tech Blog, 2017

Version B

Which gets more clicks? Vote now.

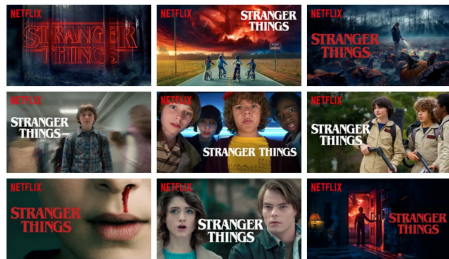
Act 1 — Netflix: *The Relatable One*

What Netflix Actually Does

- Every thumbnail you see has been **tested** against alternatives
- Conversion to “press play” varies by **up to 30%** based on image alone
- Netflix runs thousands of A/B tests simultaneously at any given time

The Lesson

Even **simple visual choices** have large, measurable effects — and **nobody can predict which version wins.**



Source: Netflix Tech Blog, 2017

Act 2 — Amazon: *The Surprising One*

Version A — original		Version B — changed
Create account to continue Email address you@email.com Password Create a password Confirm password Re-enter password Register Already have an account? Sign in Customers abandoned here	one word →	Enter your email to continue Email address you@email.com Continue You can create an account later Already have an account? Sign in Customers proceeded to checkout

“Register” → “Continue”

\$300M / year impact

The \$300M Button

- Amazon's checkout required account registration
- Designers changed one label: **“Register” → “Continue”**
- Revenue impact: **\$300 million per year**

Discussion

“Could you have predicted that?”

Think-pair-share: 60 seconds

Act 3 — Bing: *The Payoff*

The \$100M Ignored Idea

- A **junior engineer** proposed a small change to how ad headlines were displayed
- Leadership deprioritized it — too busy, seemed minor
- Engineer ran the test **anyway**
- Result: **\$100M in additional annual revenue** from a single tweak

The Organizational Lesson

“My manager has 10 years of experience and says X. A 3-week test says Y.

Which do you trust?”

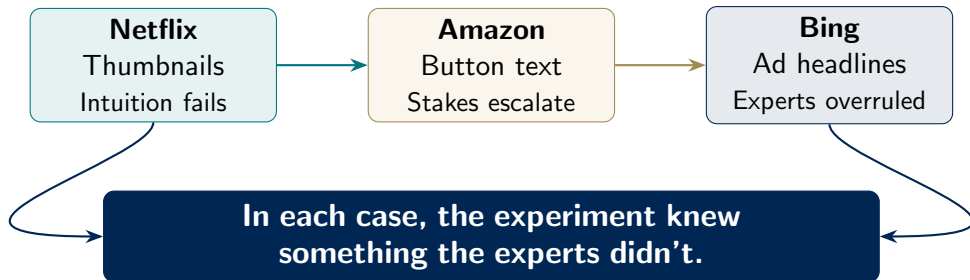
[Bing Search Ads Layout]

Before/After comparison of
the ad headline display

Annotated with \$100M revenue
uplift figure

Source: Microsoft Research blog

The Unifying Punchline



Today we learn **why** that is — and **how to run one**.

Where We've Been

Our Causal Inference Journey

Framework	Core Problem	Our Solution
Potential Outcomes	Selection Bias	Define ATE, ATT formally
DiD	Can't observe counterfactual	Parallel trends + control group
A/B Test (today)	Same problem...	Randomization

DiD Logic

*"We can't randomize, so here's how we recover causality **observationally**."*

A/B Test Logic

*"When we **CAN** randomize, here's what that buys us."*

Why Randomization Solves Selection Bias

Recall the fundamental problem:

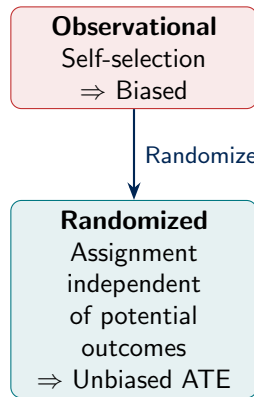
$$\underbrace{\mathbb{E}[Y_i^1 - Y_i^0]}_{\text{ATE}} \neq \underbrace{\mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0]}_{\text{Naive estimate}}$$

The gap is **selection bias**: treated units were already different.

Randomization eliminates the gap:

$$D_i \perp\!\!\!\perp (Y_i^0, Y_i^1) \Rightarrow \mathbb{E}[Y_i^0|D_i = 1] = \mathbb{E}[Y_i^0|D_i = 0]$$

$$\Rightarrow \mathbb{E}[Y_i|D_i = 1] - \mathbb{E}[Y_i|D_i = 0] = \text{ATE}$$



Key Vocabulary for Experimental Design

Assignment Mechanisms

- **Complete randomization:** Each unit assigned independently
- **Stratified randomization:** Balance on key covariates first
- **Cluster randomization:** Assign groups, not individuals

Key Estimands

- **ATE:** Average Treatment Effect

$$\tau = \mathbb{E}[Y_i^1 - Y_i^0]$$

- In a perfect RCT, the **difference in means** is an unbiased estimate of ATE

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$$

Business Translation

Treatment = new email subject line, button color, ad headline, page layout

Outcome = click-through rate, conversion, revenue per user

The Design Decision You Can't Skip

Common Mistake

"We'll run the test until we see a significant result, then stop."

This is one of the most dangerous practices in A/B testing. We will revisit it in Class 2.

What You Must Decide *Before* Running the Experiment

- ① **Minimum Detectable Effect (MDE)**: What is the smallest effect you care about?
- ② **Statistical power** ($1 - \beta$): How often do you want to detect it when it exists?
- ③ **Significance level** (α): How often can you tolerate a false positive?
- ④ **Sample size** n : Determined by the above three inputs

The Power Formula

For a two-sample test of means, the required sample size per group is:

Sample Size Formula

$$n \geq \frac{(z_{1-\alpha/2} + z_{1-\beta})^2 \cdot 2\sigma^2}{\delta^2}$$

$z_{1-\alpha/2}$ Critical value for significance (e.g., 1.96 for $\alpha = 0.05$)

$z_{1-\beta}$ Critical value for power (e.g., 0.84 for 80% power)

σ^2 Variance of the outcome variable

δ Minimum detectable effect (MDE)

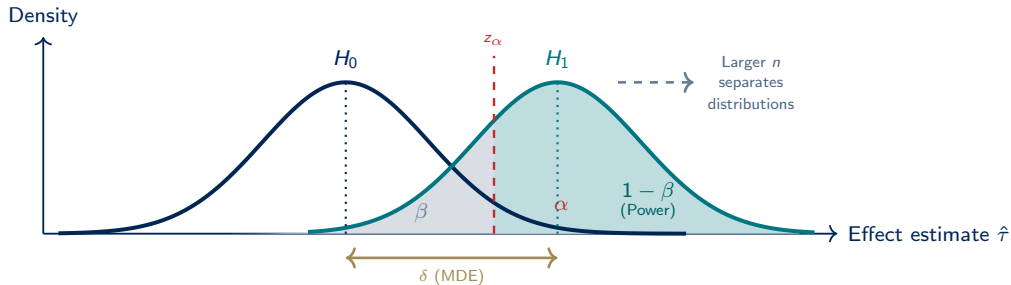
Key Insight

n grows with σ^2 , shrinks with δ^2 .
Small effects need **large** samples.

Rule of Thumb

Typical settings: $\alpha = 0.05$, power = 0.80
Industry standard: power = 0.90 for high-stakes tests

Visualizing the Four Components



Intuition: Larger $n \Rightarrow$ SE shrinks $\Rightarrow$ distributions separate $\Rightarrow$ easier to distinguish H_0 from H_1

Balance Checks: Did Randomization Work?

What Is a Balance Check?

After randomization, compare **pre-treatment covariates** across treatment and control groups. If randomization worked, these should be **statistically indistinguishable**.

Example: Email Campaign Experiment

Covariate	Treatment	Control
Avg. age	34.2	34.1
Prior purchases	2.3	2.4
Account age (mo.)	18.1	17.9
% female	52%	51%

Red Flags

- Large, statistically significant differences in *pre-treatment* variables
- This suggests the assignment mechanism was compromised

Standardized Difference

$$d = \frac{\bar{X}_T - \bar{X}_C}{\sqrt{(\hat{\sigma}_T^2 + \hat{\sigma}_C^2)/2}}$$

Rule of thumb: $|d| < 0.1$ is balanced

A Worked Example: Email Subject Line Test

The Setup

Company: Online retailer

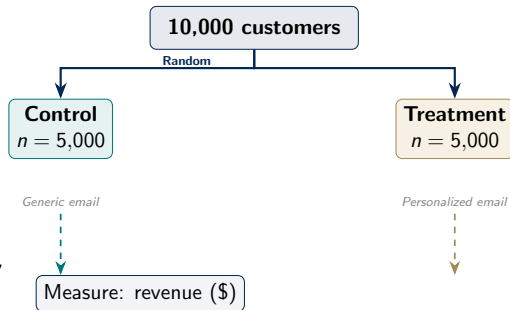
Question: Does a personalized subject line increase revenue per email sent?

Control (A) Generic: *"Shop our new arrivals"*

Treatment (B) Personal: *"Maria, your picks are in"*

Outcome: Revenue per customer (\$) in 7 days

Sample: 10,000 customers randomly split 50/50



Write your prior

Before seeing the data: how large do you expect the effect to be? Write a number.

Step 1 — Look at the Data

Summary Statistics

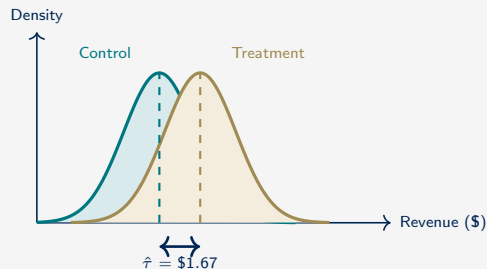
	Control (A)	Treatment (B)
n	5,000	5,000
Mean revenue $\bar{Y}$	\$12.44	\$14.11
Std. deviation s	\$18.20	\$18.65
Difference	$\hat{\tau} = \\$14.11 - \\$12.44 = \\$1.67$	

Balance Check (pre-treatment)

Covariate	Control	Treatment
Mean age	38.2	38.1
Prior purchases	3.4	3.4
Account age (mo.)	22.1	22.3

⇒ Groups look balanced. Randomization worked.

What the Distributions Look Like



Key observation

The distributions overlap massively. Is the \$1.67 gap real, or just noise?

Step 2 — Run the Test

The Two-Sample t -Test

Null hypothesis: $H_0 : \mu_B - \mu_A = 0$ (no effect)

Alternative: $H_1 : \mu_B - \mu_A \neq 0$

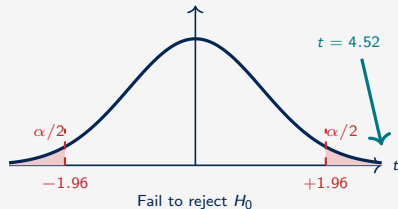
Standard error of the difference:

$$\begin{aligned} SE &= \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} = \sqrt{\frac{18.20^2}{5000} + \frac{18.65^2}{5000}} \\ &= \sqrt{0.0663 + 0.0696} = \sqrt{0.1359} \approx \mathbf{0.369} \end{aligned}$$

Test statistic:

$$t = \frac{\hat{\tau}}{SE} = \frac{1.67}{0.369} \approx \mathbf{4.52}$$

What the t -distribution tells us



Result

$$t = 4.52 \gg 1.96 \Rightarrow p < 0.001$$

We reject H_0 . The difference is statistically significant.

Step 3 — Interpret the Result

95% Confidence Interval

$$\begin{aligned}\hat{\tau} \pm 1.96 \times \text{SE} &= 1.67 \pm 1.96 \times 0.369 \\ &= 1.67 \pm 0.72 = [\$0.95, \$2.39]\end{aligned}$$

The entire interval is above zero.

We are 95% confident the true effect is between **\$0.95** and **\$2.39** per customer.

The Regression Equivalent

$$\text{Revenue}_i = 12.44 + 1.67 \cdot \text{Treatment}_i + \varepsilon_i$$

Same $\hat{\tau}$, same SE, same p -value.

This is what `statsmodels.OLS` will give you in the notebook.

Is This a Good Result?

Metric	Value
Effect size $\hat{\tau}$	\$1.67 per customer
95% CI	[\$0.95, \$2.39]
p -value	< 0.001
Relative lift	+13.4%
Emails sent / month	500,000
Est. monthly gain	≈ \$835,000

Statistical vs. Economic Significance

$p < 0.001$: statistically significant ✓

\$1.67 per customer at scale: economically significant ✓

Both questions must be answered before you ship the change.

Notebook Exercise: Simulating a Perfect RCT

① Write your prior first

In the notebook, a cell asks: *“If the true email click-through effect is +2 pp, what do you expect the difference-in-means to estimate?”* Write a number before running anything.

② Simulate the experiment

Run the calibrated simulation cell. True effect = `TRUE_EFFECT` = 2.0. Recover the ATE using difference-in-means.

③ Introduce selection bias

Modify the assignment so that high-intent users are more likely to get treatment. Re-estimate. How biased is the result?

④ Run a balance check

Compute standardized differences before and after biased assignment.

Class 1 Wrap-Up

What We Covered

- Three real business experiments that beat expert intuition
- Randomization as the solution to selection bias (in potential outcomes notation)
- Power, sample size, MDE tradeoffs
- Balance checks as verification

Teaser for Class 2

Randomization **solves** selection bias — but only when it works perfectly.

What happens when:

- Not everyone who is assigned treatment *takes* it?
- Some people drop out of the experiment?
- Researchers run 50 tests and report only the one that worked?

Class 2

Where Experiments Break

Non-compliance, p-hacking, and real-world applications

Difference-in-Means = OLS

A Result Worth Knowing

The **difference-in-means estimator** is algebraically identical to the OLS coefficient on a binary treatment indicator:

$$Y_i = \alpha + \tau D_i + \varepsilon_i \quad \Rightarrow \quad \hat{\tau}^{\text{OLS}} = \bar{Y}_1 - \bar{Y}_0 = \hat{\tau}^{\text{ATE}}$$

Why This Matters

- You already know how to run this!
- Adds **standard errors, confidence intervals**, and **p-values** automatically
- Easy to extend: add covariates to **reduce variance** and improve precision

Extended Regression (ANCOVA)

$$Y_i = \alpha + \tau D_i + \beta X_i + \varepsilon_i$$

Adding pre-treatment controls X_i doesn't change $\hat{\tau}$ in expectation, but **reduces standard error** — same precision with smaller n .

When Not Everyone Complies

A Very Common Real-World Problem

In business experiments, users *assigned* to treatment may not actually *receive* it.

Examples:

- Assigned to see new checkout flow, but didn't visit checkout
- Assigned to receive promotional email, but it bounced / went to spam
- Assigned to new UI, but used the mobile app instead (which wasn't updated)

Two Types of Non-Compliers

Never-takers Assigned to treatment, never take it

Always-takers Assigned to control, somehow get treatment

The Problem

If you compare *actual takers* vs. *non-takers*, you reintroduce selection bias — because takers and non-takers differ systematically!

ITT vs. LATE — Two Estimands

Intention-to-Treat (ITT)

Compare based on **assignment**, regardless of actual treatment receipt.

$$ITT = \mathbb{E}[Y_i | Z_i = 1] - \mathbb{E}[Y_i | Z_i = 0]$$

Unbiased and **conservative**.

Answers: “What is the effect of *offering* the treatment?”

Local Average Treatment Effect (LATE)

The treatment effect for **Compliers** only — those who take treatment when assigned and don't when not assigned.

$$LATE = \frac{ITT}{\text{Compliance Rate}}$$

Estimated via **Instrumental Variables** (assignment Z instruments for treatment D).
Answers: “What is the effect for those who actually respond to the assignment?”

Business Context

Email campaign: ITT = “effect of sending the email.” LATE = “effect of *opening* the email

[Compliance Type Diagram]

2×2 table: rows = assignment ($Z = 0, 1$), columns = treatment received ($D = 0, 1$)

Label each cell:

- ($Z = 1, D = 1$): Compliers + Always-takers
 - ($Z = 1, D = 0$): Never-takers
- ($Z = 0, D = 0$): Compliers + Never-takers
 - ($Z = 0, D = 1$): Always-takers (defiers)

Highlight the Complier group with the LATE arrow

Generate as diagram in companion notebook

The Multiple Testing Problem

The Danger: Testing Until Significant

If you run 20 independent tests with $\alpha = 0.05$, the probability of **at least one false positive** is:

$$1 - (1 - 0.05)^{20} \approx \mathbf{64\%}$$

This is not statistical power. This is **luck**.

How P-hacking Happens in Business

- Running the test until $p < 0.05$ appears (“peeking”)
- Testing 10 metrics, reporting only the significant one
- Subsetting data until a subgroup shows significance

Corrections

Bonferroni $\alpha^* = \alpha/m$ (conservative)

Benjamini-Hochberg Controls *false discovery rate* (less conservative)

Pre-registration Declare hypotheses before

What P-hacking Looks Like

[P-value Distribution Under Null Hypothesis]

Two side-by-side histograms of p -values:

Left: “Honest” testing — p -values uniform under H_0

Right: P-hacking — spike near $p = 0.05$ from selective reporting

Red vertical line at $p = 0.05$

Reproduce in companion notebook with simulated tests

A suspicious spike just below $p = 0.05$ in published research is called the “**p-curve.**” It is evidence of selective reporting.

A Second Failure Mode: Attrition

What Is Attrition?

Some participants **drop out** of the experiment before the outcome is measured. This is fine if dropout is **random**. It is a problem if dropout is **related to treatment**.

Business Example

Experiment: New onboarding flow

Outcome: Purchase within 30 days

Treatment group has **higher dropout** within the first 3 days

⇒ The remaining treatment users are unusually committed

⇒ Comparing them to full control is biased

Attrition Bias Direction

If treatment causes *high-outcome types* to stay and *low-outcome types* to drop out, you will **overestimate** the treatment effect.

Always check: is attrition **balanced** across treatment and control?

A Field Experiment in Business: Lyst E-commerce Study

[Case Study Screenshot / Paper Cover]

Image of the experiment design
(product page layout A vs. B)

*Instructor: choose a published
business RCT or use the Booking.com
"Controlled Experiments at Scale" paper*

What Good Business A/B Testing Looks Like

- Pre-registered hypothesis and sample size
- Balance check reported in paper/deck
- ITT as primary estimate
- Multiple testing correction applied
- Attrition checked and minimal
- Economic, not just statistical, significance emphasized

Discussion Question

*"Statistical significance at $p < 0.05$ with
 $n = 2,000,000$*

Statistical vs. Economic Significance

The Key Distinction

With very large samples, even tiny effects become statistically significant.

A button color change that increases clicks by **0.001%** can have $p < 0.001$ with $n = 10^7$.

Always Ask Two Questions

- 1 Is the effect statistically significant?
- 2 Is the effect **large enough to matter** for business decisions?

The Bayesian Investor Framing

Before you run an experiment, ask:
“What effect size would change my decision?”

This is your **minimum detectable effect** and your **economic significance threshold** — and they should be the same number.

Break the Estimator

What Goes Wrong When Experiments Break?

In the notebook, we will systematically **violate assumptions** and observe the consequences.

Violation 1: Non-compliance

- Introduce non-compliance (50% of treated units don't take treatment)
- Compare: naive, ITT, LATE
- Observe: ITT is conservative; naive is biased

Violation 2: P-hacking

- Run 100 tests on pure noise

Violation 3: Attrition Bias

- Make high-intent users drop out of control
- Re-estimate ATE
- Observe: upward bias in effect estimate

Expected Results Table

Scenario	True	Estimated
Clean RCT	2.0	≈ 2.0

Notebook Exercise: Breaking the Experiment

① Write your prior first

Predict the estimated effect under each of the three violation scenarios. Fill in the “Your Prior” column before running any code.

② Introduce non-compliance

Modify the assignment variable so 40% of treated units don't comply. Compute ITT, then use the compliance rate to back out LATE.

③ Simulate p-hacking

Run a loop: 100 tests on null data, record p -values. Count false positives. Apply Bonferroni correction and recount.

④ Introduce attrition

Drop the bottom 30% of potential outcomes from the control group. Re-estimate. Which direction does the bias go?

Bridge to Text & Unstructured Data

Where A/B Testing Meets Text

Most business experiments **generate** unstructured data:

- Open-ended survey responses: *“Why did you not complete the purchase?”*
- Customer reviews triggered by a new feature
- Support tickets from users exposed to treatment
- Social media sentiment shifts post-experiment

The Question for Next Week

How do we measure **text outcomes**?

How do we know if treatment *changed how people*

[Pipeline Diagram]

A/B Test → Outcome includes text

Text → Sentiment score / Topic model
→ Numeric outcome

Then apply the same causal framework

*Simple flowchart connecting
RCT framework to text analysis*

Two-Class Summary

Class 1 Key Takeaways

- Business experiments routinely outperform expert intuition
- Randomization eliminates selection bias by making $D_i \perp (Y_i^0, Y_i^1)$
- Power analysis must come **before** data collection
- Balance checks verify randomization worked

Class 2 Key Takeaways

- Non-compliance: ITT is safe; LATE uses IV
- Multiple testing inflates false positive rates dramatically
- Attrition bias is directional and diagnosable
- Statistical significance $\neq$ economic significance

The Through-Line

Randomization is the **gold standard** for causal inference.

But experiments fail for predictable, diagnosable reasons.

Knowing the failure modes is as important as knowing the method

References & Further Reading

Core Readings

- Cunningham, *Causal Inference: The Mixtape*, Ch. 4 (Potential Outcomes review) and Ch. 7 (Instrumental Variables for LATE)
- Duflo, Glennerster & Kremer (2007), “Using Randomization in Development Economics Research”
- Kohavi, Tang & Xu (2020), *Trustworthy Online Controlled Experiments* (O'Reilly)

Case Studies

- Microsoft Bing headline test (Kohavi et al.)
- Amazon “Register” button (*The \$300M Button*, Luke Wroblewski)
- Netflix thumbnail A/B testing (Netflix Tech Blog)
- Booking.com controlled experiments at scale (Ron Kohavi)

Questions?

“The experiment doesn’t lie. The question is whether you ran it correctly.”

Open the companion notebook and let’s code.